

Context utility depends on its readout

Central finding: during matched training, native inference reverses its preference for support context while a fixed prototype does not. The end-to-end effect of graph context therefore cannot be read as the usefulness of the information it encodes.

A A real but non-universal phenomenon

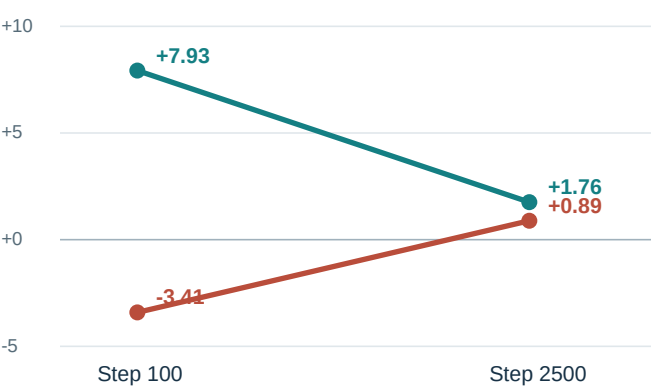
Support suppression: pooled AUC change, points

	Pol.	Elect.	Pages	Bots	Susp.
COVID	-0.1*	+0.0*	+0.1	-4.0	+1.2
Political	-5.7	-0.4	-3.9	-2.4	-2.8
Hong Kong	+9.9	+2.0	+2.5	-7.5	+0.6*
Election	-0.2	-0.0*	-1.3	-10.1	+0.2*
Pages	+1.8	+1.3	+0.9	-1.6	+1.4
Midterm	+3.1	+0.3	+0.8	-0.6	-0.1*
Bots	+0.8	+0.1	+0.1*	-2.6	+0.2*
Ukraine	-0.1	-0.0	-0.0*	-5.4	+1.7
Suspended	+7.4	+0.1*	-0.6	-8.3	+1.4

Nine sources, five targets; mean of two episode streams, not training seeds. Teal helps, red hurts; * signs differ. Hong Kong is an extreme, not the only affected source.

B Training reverses only the native preference

Page suppression gain: within-episode AUC points



Same page episodes, 3 seeds x 2 streams; mean effects. Prototypes improve with suppression in all six comparisons at both steps. Native: 5/6 harmed early, 6/6 helped later. Both improve absolutely. **The suppressed path catches up; intact inference does not deteriorate.**

C Training utility is not inference utility

Fixed four-target mean AUC; both episode streams

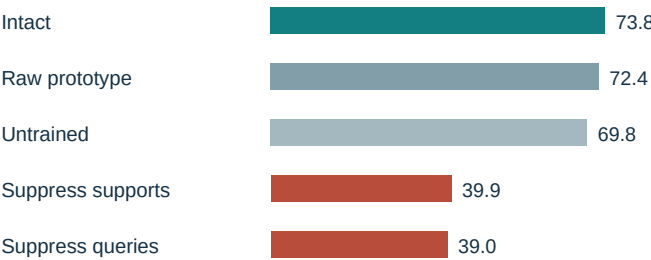
Training / readout	Blocked	Interlv.
Native / full	0.831	0.820
Native / ridge	0.847	0.842
Ridge-only / ridge	0.809	0.811
Isolated / full	0.800	0.801

Matched initialization and per-source examples; one seed, two schedules. Native training yields stronger ridge representations than ridge-only on 3/4 blocked and 4/4 interleaved targets in both streams. **Isolation fails the prespecified repair gate.**

Counterexample: on blocked political classification, full inference beats its ridge readout. The solver is not uniformly harmful. Encoder trajectories diverged; this is not a same-encoder causal comparison.

D Generality result: both roles can help

Public Wiki to FB15K-237: accuracy (%)



A useful original-style checkpoint needs both roles: support/query suppression costs 33.87/34.80 points. 500 paired 20-way episodes, 8,001 updates. This contradicts universal support harm. Public queries are not fixed under the role intervention; do not claim the social mediation result generalizes.

Contribution: matched graph-role interventions show why native context utility must be distinguished from information available to a fixed readout. At page step 100, suppressed prototype AUC is .649 versus native .504; the information exists but native inference does not exploit it. This extends beyond a static graph-distance account, without claiming a universal predictor or isolating one defective projection.

Decision: pursue the explanation, not an isolation method. The completed training test contradicts the simple remedy of blocking native gradients. No three-seed method expansion. The precise cause of the readout mismatch remains unresolved; query processing differs too. This is a contribution candidate, not an established mechanism.

Evidence: page prototype (8ccd3686); nine-source map; isolation training/evaluation (0b9d9641); public quality/roles. Public logit-tolerance replay failed; an explicit amendment before scientific outcomes required exact labels and all 40,000 predicted classes, which matched. No tolerances were enlarged. These are conditional results, not seed-level generality.